

Consideration on the learning efficiency of multiple-layered neural networks with linear units

Miki Aoyagi

*College of Science & Technology, Nihon University, 1-8-14, Surugadai, Kanda,
Chiyoda-ku, Tokyo 101-8308 Japan*

Abstract

In the last two decades, remarkable progress has been done in singular learning machine theories on the basis of algebraic geometry. These theories reveal that we need to find resolution maps of singularities for analysing asymptotic behaviour of state probability functions when the number of data increases. In particular, it is essential to construct normal crossing divisors of average log loss functions. However, there are few examples for obtaining these for singular models. In this paper, we determine the resolution map and normal crossing divisors for multiple-layered neural networks with linear units. Moreover, we have the exact values for the learning efficiency, which is so called learning coefficients. Multiple-layered neural networks with linear units are simple, however, very important models because these models give the essential information from data of input-output pairs. Moreover, these models are very close to multiple-layered neural networks with rectified linear units (ReLU). We show the learning coefficients of multiple-layered neural networks with linear units are bounded even though the number of layers goes to infinity, which means that the main term of asymptotic expansion of the free energy and generalization error of singular models are much smaller than the dimension of its parameter space.

Keywords: resolution map, singular learning theory, multiple-layered neural networks with linear units, algebraic geometry

*Corresponding author

Email address: aoyagi.miki@nihon-u.ac.jp (Miki Aoyagi)

1. Introduction

Singular learning models include hierarchical learning machines such as neural networks, reduce rank regression, Boltzmann machine, normal mixture models and so on. Even though these models have been widely used to analyse real data, theoretical analysis have not been developed sufficiently. These models have non-positive fisher information matrices and therefore cannot be analysed by classical theories [1].

For example, let us consider the reduce rank regression model, which is one of singular models. Let

$$\{w = (A^{(1)}, A^{(2)}) \mid A^{(s)} \text{ is an } H^{(s)} \times H^{(s+1)} \text{ matrix } \},$$

be the set of parameters. Assume that the input value is a vector $X \in \mathbf{R}^{H^{(3)}}$ with probability density function $r(X)$ and output value a vector $Y \in \mathbf{R}^{H^{(1)}}$ given by

$$Y = A^{(1)}A^{(2)}X + (\text{noise}),$$

with Gaussian noise. Then, the statistical learning model is obtained by

$$pl(X, Y|w) = \frac{1}{(\sqrt{2\pi})^{H^{(1)}}} \exp(-\frac{1}{2}\|Y - A^{(1)}A^{(2)}X\|^2)r(X).$$

This model has $H^{(3)}$ input units, $H^{(1)}$ output units, and $H^{(2)}$ hidden units. Let us assume that the $w^* = (A^{*(1)}, A^{*(2)})$ be the true parameter, that is, we have the true probability density function

$$r(X, Y) = \frac{1}{(\sqrt{2\pi})^{H^{(1)}}} \exp(-\frac{1}{2}\|Y - A^{*(1)}A^{*(2)}X\|^2)r(X).$$

Let $(X, Y)^n := \{(X_i, Y_i)\}_{i=1}^n$ be n training samples selected independently and identically from $r(X, Y)$.

Define the average log loss function $L(w)$ by $L(w) = -E_{X,Y}[\log pl(X, Y|w)]$ and the set of optimal parameters W_0 by

$$W_0 = \{w_0 \in W \mid L(w_0) = \min_{w' \in W} L(w')\} = \{(A^{(1)}, A^{(2)}) \mid A^{*(1)}A^{*(2)} = A^{(1)}A^{(2)}\}.$$

10 Because $r(X, Y) = pl(X, Y|(A^{*(1)}, A^{*(2)}))$, we have $r(X, Y) = pl(X, Y|w_0)$ for all $w_0 \in W_0$.

If the rank of $A^{*(1)}A^{*(2)}$ is less than $\min\{H^{(1)}, H^{(2)}, H^{(3)}\}$, then W_0 is not one point, but its dimension is positive. It is obvious that the Fisher matrix function $(\frac{\partial^2 L(w)}{\partial w_i \partial w_j})$ at $w^* = (A^{*(1)}, A^{*(2)})$ has zero eigenvalues, therefore, we
 15 cannot apply classical theories, for example, model selection methods AIC [2], TIC [3], HQ [4], NIC [5], BIC [6], MDL [7], because their methods need regular conditions such as a positive definite Fisher information matrix and the unique point minimizing a log loss function.

Let $\varphi(w)$ be an *a priori* probability density function on a parameter set W and $ps_n(w)$ be the *a posteriori* probability density function,

$$ps_n(w) = \frac{1}{\int_W \varphi(w) \prod_{i=1}^n pl(X_i, Y_i|w) dw} \varphi(w) \prod_{i=1}^n pl(X_i, Y_i|w).$$

We then have the predictive density function in Bayesian estimation

$$pr_n(X, Y) = \int pl(X, Y|w) ps_n(w) dw.$$

Let G_n be the Bayes generalization loss,

$$\begin{aligned} G_n &= - \int r(X, Y) \log pr_n(X, Y) dX dY \\ &= - \int r(X, Y) \log r(X, Y) dX dY + \int r(X, Y) \log \frac{r(X, Y)}{pr_n(X, Y)} dX dY \\ &= L(w_0) + E_{X,Y}[\log \frac{r(X, Y)}{pr_n(X, Y)}]. \end{aligned}$$

$E_{X,Y}[\log \frac{r(X, Y)}{pr_n(X, Y)}]$ is the Kullback function, which is the distance between $r(X, Y)$ and $pr_n(X, Y)$. Watanabe [8, 9, 10, 11] proved the following relation,

$$E[G_n] = L(w_0) + \frac{1}{n} \lambda + o(\frac{1}{n}),$$

20 where λ is the learning coefficient, which was obtained in Theorem 1 [12]. These values are much smaller than the dimension $H^{(1)}H^{(2)} + H^{(2)}H^{(3)}$ of parameter spaces. This fact means learning efficiencies are increased more than regular models, as a model's complexity increases.

Based on the free energy,

$$\begin{aligned} F_n(\beta) &= -\frac{1}{\beta} \log \int \prod_{i=1}^n pl(X_i, Y_i|w)^\beta \varphi(w) dw \\ &= nL_n(w_0) + \frac{\lambda}{\beta} \log(n) - \frac{\theta-1}{\beta} \log \log(n) + o_p(1) \end{aligned}$$

for inverse temperature $\beta > 0$, which was shown by Watanabe [1], we have the
 25 model-selection method, “singular Bayesian information criterion” (sBIC [13]).
 sBIC uses the learning coefficients λ and its order θ very effectively with a
 fix point equation system of marginal likelihoods. In addition, the learning
 efficiency λ and its order θ of a leaning model give mathematical indicators
 30 for analysing and developing the precision of numerical methods, such as the
 widely-applicable information criterion WAIC [2, 8, 9, 14, 1, 10, 11] and cross-
 validation in the Bayesian estimation approach model selection methods.

The learning efficiency λ is equal to a log canonical threshold of a Kullback
 function in algebraic geometry whose exact value is obtained using recursive
 35 blow-up process [15]. However, it is difficult to obtain these thresholds, be-
 cause we need to determine all branches of its blow-up process to obtain nor-
 mal crossing divisors and Kullback functions degenerate with respect to their
 Newton polyhedra [16]. Moreover, theories for log canonical thresholds are
 not fruitful for the real field, compared with the complex field or algebraically
 40 closed fields [17, 18, 19]. Therefore, it is of interest in various fields, even in
 mathematics, to obtain these thresholds. In recent studies, we obtained exact
 values or bounded values of the learning coefficients for Vandermonde matrix-
 type singularities, which are related to the three-layered neural networks and
 normal mixture models, among others [20, 21, 22, 23, 24]. We have also ex-
 45 act values for the restricted Boltzmann machine [25]. Additionally, Rusakov
 and Geiger [26, 27], Zwiernik [28] and Drton et al. [29] respectively, obtained
 these coefficients for naive Bayesian networks, directed tree models with hidden
 variables, and the Gaussian latent tree and forest models.

Our purpose in this paper is to obtain λ and θ for multiple-layered neural
 50 networks with linear units.

In Section 2, we overview the definitions of resolution of singularities and log canonical thresholds, and we show our main results in Section 3. In Section 4, the main theorem's proof is obtained, and we conclude in Section 5.

2. Resolution of singularities and log canonical threshold

55 We denote constants by superscript $*$, for example, a^* , b^* , and w^* .

Definition 1. *The log canonical threshold for an analytic function F and a C^∞ function with a compact support in a neighbourhood U of w^* , is defined as*

$$\lambda_{w^*}(F, \varphi) = \sup\{c : \int_U |F|^{-kc} \varphi(w) dw < \infty\},$$

where $k = 1$ over the real field and $k = 2$ over the complex field. The value $\lambda_{w^*}(F, \varphi)$ is equal to the largest pole of the zeta function $\int_U |F|^{kz} \varphi(w) dw$ for $z \in \mathbf{C}$, respectively. Also let $\theta_{w^*}(F, \varphi)$ be its order.

If $\varphi(w^*) \neq 0$, then denote $\lambda_{w^*}(F) = \lambda_{w^*}(F, \varphi)$ and $\theta_{w^*}(F) = \theta_{w^*}(F, \varphi)$ 60 because the log canonical threshold and its order are independent of φ .

For ideal J , generated by real analytic functions $F_1, \dots, F_m$ in a neighbourhood of w^* , define $\lambda_{w^*}(J) = \lambda_{w^*}(F_1^2 + \dots + F_m^2)$.

Here, $\lambda_{w^*}(J)$ for ideal J is well-defined by Lemma 1.

Lemma 1 ([30, 31, 32]). *Let $\mathcal{J} = \langle F_1, \dots, F_n \rangle$ be the ideal generated by analytic functions $F_1, \dots, F_n$ on a neighbourhood U of $w^* \in \mathbf{R}^d$. Also, let $G_1, \dots, G_m$ 65 be analytic functions on U .*

We have

$$\lambda_{w^*}(G_1^2 + \dots + G_m^2) \leq \lambda_{w^*}(F_1^2 + \dots + F_n^2),$$

if $G_1, \dots, G_m \in \mathcal{J}$.

In particular, if $\mathcal{J} = \langle G_1, \dots, G_m \rangle$, then we have

$$\lambda_{w^*}(F_1^2 + \dots + F_n^2) = \lambda_{w^*}(G_1^2 + \dots + G_m^2).$$

Definition 2. *Let $C = (c_{ij})$ be a matrix. Define $\|C\|$ as the norm of C by $\|C\| = \sqrt{\sum_{i,j} |c_{ij}|^2}$. Also define $\langle C \rangle$ as the ideal generated by all elements c_{ij} 70 of C .*

Applying Hironaka's Theorem [15] to Kullback functions

$$K(w) = E_{X,Y}[\log \frac{r(X,Y)}{pl(X,Y|w)}] = \int r(X,Y) \log \frac{r(X,Y)}{pl(X,Y|w)} dXdY,$$

where $r(X,Y)$ is the true probability density function and $pl(X,Y|w)$ is a learning model with a parameter $w \in W$, we obtain the proper analytic map π from a manifold Q to neighbourhood $V \subset W$,

$$K(\pi(u)) = u_1^{2k_1} u_2^{2k_2} \cdots u_d^{2k_d},$$

$$\pi'(u)\varphi(\pi(u)) = u_1^{h_1} u_2^{h_2} \cdots u_d^{h_d},$$

where $\varphi(w)$ is an a priori density function, $(u_1, \dots, u_d)$ is a local analytic coordinate system on $U \subset Q$, and $k_1, \dots, k_d, h_1, \dots, h_d$ are non-positive integers. Then, we have

$$\int_V |K(w)|^{-c} \varphi dw = \int_Q |K(g'(u))|^{-c} \varphi(\pi(u)) \pi'(u) du.$$

Therefore, the log canonical threshold, i.e., the learning coefficient is

$$\lambda = \min_U \min_{1 \leq j \leq d} \frac{h_j + 1}{2k_j},$$

and its order

$$\theta = \max_u \text{Card}\{j : \frac{h_j + 1}{2k_j} = \lambda\},$$

where $\text{Card}(S)$ denotes the cardinality of a set S .

3. Multiple-layered neural networks with linear units

In the paper [12], the learning coefficients for the reduced rank regression model, i.e., the three-layered neural networks with linear units were obtained.

Theorem 1 ([12]). *We will use the same notation as in Section 1. Assume that the a priori probability density function $\varphi(w)$ is a C^∞ -function with compact support W , satisfying $\varphi(A^{*(1)}, A^{*(2)}) > 0$.*

λ and θ for the model corresponding to the log canonical threshold

$$\lambda_{(A^{*(1)} A^{*(2)})}(\|A^{(1)} A^{(2)} - A^{*(1)} A^{*(2)}\|^2)$$

and its order θ are as follows.

Let r be the rank of $A^{*(1)}A^{*(2)}$ and $M^{(s)} = H^{(s)} - r$. Additionally, let

$$\mathcal{M} = \begin{cases} \{1, 2, 3\}, & \text{if } M^{(s)} < \frac{\sum_{i=1}^3 M^{(i)}}{2} \text{ for } s = 1, 2, 3, \\ \{1, 2\}, & \text{if } M^{(3)} \geq M^{(1)} + M^{(2)}, \\ \{1, 3\}, & \text{if } M^{(2)} \geq M^{(1)} + M^{(3)}, \\ \{2, 3\}, & \text{if } M^{(1)} \geq M^{(2)} + M^{(3)}, \end{cases}$$

$$\ell = \text{Card}(\mathcal{M}) - 1,$$

$$\tilde{M} = \frac{\sum_{i \in \mathcal{M}} M^{(i)}}{\ell},$$

$$M \in \mathbf{N} \text{ such that } M - 1 < \tilde{M} \leq M,$$

and

$$a = \sum_{i \in \mathcal{M}} M^{(i)} - (M - 1)\ell.$$

Then, we have

$$\begin{aligned} \lambda = & \frac{-r^2 + r(H^{(1)} + H^{(3)})}{2} + \frac{a(\ell - a)}{4\ell} \\ & - \frac{\ell(\ell - 1)}{4} \left(\frac{\sum_{i \in \mathcal{M}} M^{(i)}}{\ell} \right)^2 + \frac{1}{2} \sum_{i < j, i, j \in \mathcal{M}} M^{(i)} M^{(j)} \end{aligned}$$

and

$$\theta = a(\ell - a) + 1.$$

80 4. Main Theorem

Define matrices $A^{(s)}$ of size $H^{(s)} \times H^{(s+1)}$ for $s = 1, \dots, L$,

$$A^{(s)} = (a_{ij}^{(s)}), \quad (1 \leq i \leq H^{(s)}, 1 \leq j \leq H^{(s+1)}).$$

Let W be the set of parameters

$$W = \{w = \{A^{(s)}\}_{1 \leq s \leq L} \mid A^{(s)} \text{ is an } H^{(s)} \times H^{(s+1)} \text{ matrix}\}.$$

Denote the input value by $X \in \mathbf{R}^{H^{(L+1)}}$ with probability density function $r(X)$ and output value $Y \in \mathbf{R}^{H^{(1)}}$ for the multiple-layered neural network with linear units, which is given by

$$Y = \prod_{s=1}^L A^{(s)} X + (\text{noise}),$$

with Gaussian noise. Consider the statistical model

$$pl(Y|X, w) = \frac{1}{(\sqrt{2\pi})^{H^{(1)}}} \exp(-\frac{1}{2} \|Y - \prod_{s=1}^L A^{(s)} X\|^2), \quad pl(X, Y|w) = pl(Y|X, w)r(X).$$

The model has $H^{(1)}$ input units, $H^{(L+1)}$ output units, and $H^{(s)}$ hidden units in each hidden layer. Let

$$w^* = \{A^{*(s)}\}_{1 \leq s \leq L},$$

be the true parameter. Assume that the true density function

$$r(Y|X) = \frac{1}{(\sqrt{2\pi})^{H^{(1)}}} \exp(-\frac{1}{2} \|Y - \prod_{s=1}^L A^{*(s)} X\|^2), \quad r(X, Y) = r(Y|X)r(X),$$

which is included in the learning model. Moreover, assume that the *a priori* probability density function $\varphi(w)$ is a C^∞ -function with compact support W , satisfying $\varphi(w^*) > 0$. Then, λ and θ for the model corresponding to the log canonical threshold $\lambda_{w^*}(\|\prod_{s=1}^L A^{(s)} - \prod_{s=1}^L A^{*(s)}\|^2)$ and its order θ are as follows.

Definition 3. Let $M^{(s)} = H^{(s)} - r$ for $s = 1, \dots, L+1$.

Define $\mathcal{M} = \{M^{(S_j)} : j = 1, \dots, \ell+1\} \subset \{M^{(s)} \mid s = 1, \dots, L+1\}$ such that

$$M^{(S_j)} < M^{(s)} \text{ for } M^{(s)} \notin \mathcal{M}$$

and

$$\begin{aligned} \sum_{k=1}^{\ell+1} M^{(S_k)} &> \ell M^{(s)} \text{ for } M^{(s)} \in \mathcal{M} \\ \sum_{k=1}^{\ell+1} M^{(S_k)} &\leq (\ell-1) M^{(s)} \text{ for } M^{(s)} \notin \mathcal{M}. \end{aligned}$$

Let M be the integer such that

$$M - 1 < \frac{\sum_{k=1}^{\ell+1} M^{(S_k)}}{\ell} \leq M,$$

and

$$a = \sum_{k=1}^{\ell+1} M^{(S_k)} - (M - 1)\ell.$$

Theorem 2. *We have*

$$\begin{aligned} \lambda &= \frac{-r^2 + r(H^{(1)} + H^{(L+1)})}{2} + \frac{a(\ell - a)}{4\ell} \\ &\quad - \frac{\ell(\ell - 1)}{4} \left(\frac{\sum_{j=1}^{\ell+1} M^{(S_j)}}{\ell} \right)^2 + \frac{1}{2} \sum_{1 \leq i < j \leq \ell+1} M^{(S_i)} M^{(S_j)} \\ &= \frac{-r^2 + r(H^{(1)} + H^{(L+1)})}{2} \\ &\quad + \frac{a(\ell - a)}{4\ell} - \frac{\ell(\ell - 1)}{4} \left(M + \frac{a - \ell}{\ell} \right)^2 + \frac{1}{2} \sum_{1 \leq i < j \leq \ell+1} M^{(S_i)} M^{(S_j)} \\ &= \frac{-r^2 + r(H^{(1)} + H^{(L+1)})}{2} - \frac{1}{4} (\ell - a - 1)(\ell - a) \\ &\quad - \frac{\ell(\ell - 1)}{4} \left(M^2 + 2 \frac{a - \ell}{\ell} M \right) + \frac{1}{2} \sum_{1 \leq i < j \leq \ell+1} M^{(S_i)} M^{(S_j)} \end{aligned}$$

and

$$\theta = a(\ell - a) + 1.$$

Example If $M^{(1)} = M^{(2)} = \dots = M^{(L+1)}$, then, we have $\ell = L$, the integer M such that $M - 1 < \frac{L+1}{L} M^{(1)} \leq M$ and $a = (L + 1)M^{(1)} - (M - 1)L$.

We have

$$\lambda = \frac{-r^2 + 2rH^{(1)}}{2} + \frac{a(L - a)}{4L} + \frac{L + 1}{4L} (M^{(1)})^2$$

and

$$\theta = a(L - a) + 1.$$

Figure 1 shows that the curve of λ and θ , when $r = 0$, $M^{(1)} = 100$ and $L = 2, \dots, 100$ in Example 1.

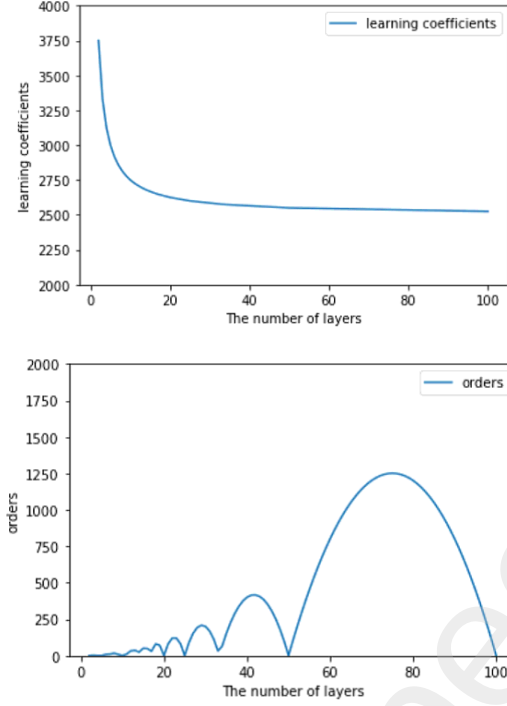


Figure 1: The curves of λ and θ , when $r = 0$, $M^{(1)} = 100$ and $L = 2, \dots, 100$ in Example 1

95 5. Proof of Main Theorem

Define E_s be the identity matrix of size s .

Lemma 2. Let $A = (a_{ij})$ be an $h_1 \times h_2$ matrix of variables with rank r_1 . Let $r \leq r_1$ and define $A = \begin{pmatrix} A_1 & A_2 \\ A_3 & A_4 \end{pmatrix}$, where A_1 is a regular matrix of size $r \times r$.

Then, there exist regular matrices $Q_1 = \begin{pmatrix} E_r & O \\ F_3 & E_{h_1-r} \end{pmatrix}$ and $Q_2 = \begin{pmatrix} E_r & F_2 \\ O & E_{h_2-r} \end{pmatrix}$ such that

$$Q_1 A Q_2 = \begin{pmatrix} A_1 & O \\ O & C_4 \end{pmatrix},$$

with $\text{rank}(C_4) = r_1 - r$.

100 (Proof)

Because A_1 is regular, let $Q_1 = \begin{pmatrix} E_r & O \\ -A_3A_1^{-1} & E_{h_1-r} \end{pmatrix}$. Then,

$$Q_1A = \begin{pmatrix} A_1 & A_2 \\ O & -A_3A_1^{-1}A_2 + A_4 \end{pmatrix}.$$

We transform the variables A_4 to $C_4 = -A_3A_1^{-1}A_2 + A_4$, to obtain

$$Q_1A = \begin{pmatrix} A_1 & A_2 \\ O & C_4 \end{pmatrix}.$$

Let $Q_2 = \begin{pmatrix} E_r & -A_1^{-1}A_2 \\ O & E_{h_2-r} \end{pmatrix}$, then we have $Q_1AQ_2 = \begin{pmatrix} A_1 & O \\ O & C_4 \end{pmatrix}$.

We transform the variables A_2 and A_3 to $F_2 = -A_1^{-1}A_2$ and $F_3 = -A_3A_1^{-1}$, respectively. Then, we have

$$\begin{aligned} Q_1 &= \begin{pmatrix} E_r & O \\ F_3 & E_{h_1-r} \end{pmatrix} \\ Q_2 &= \begin{pmatrix} E_r & F_2 \\ O & E_{h_2-r} \end{pmatrix}, \end{aligned}$$

105 which completes the proof.

End of Proof.

Let the matrix $A^{(s)}$ be in a neighbourhood of $A^{*(s)}$ of rank $r^{(s)}$. In considering $\prod_{s=1}^L A^{(s)}$ in a neighbourhood of a matrix of rank r , we have $r^{(s)} \geq r$. By Lemma 1, we can assume $\prod_{s=1}^L A^{*(s)} = \begin{pmatrix} E_r & O \\ O & O \end{pmatrix}$.

Theorem 3. *There exist regular matrices $P_1 = \begin{pmatrix} E_r & O \\ F_3 & E_{H^{(1)}-r} \end{pmatrix}$ and $P_2 =$*

$\begin{pmatrix} E_r & F_2 \\ O & E_{H^{(L+1)}-r} \end{pmatrix}$ such that

$$P_1\left(\prod_{s=1}^L A^{(s)}\right)P_2 = \begin{pmatrix} C_1 & O \\ O & \prod_{s=1}^L C^{(s)} \end{pmatrix}.$$

110 (Proof)

Let $A^{(1)} = \begin{pmatrix} A_1^{(1)} & A_2^{(1)} \\ A_3^{(1)} & A_4^{(1)} \end{pmatrix}$, where $A_1^{(1)}$ is a matrix of size $r \times r$. Because $\prod_{s=1}^L A^{*(s)} = \begin{pmatrix} E_r & O \\ O & O \end{pmatrix}$, we can assume that $\text{rank}(A_1^{(1)}) = r$. By

Lemma 2, there exist regular matrices $Q_1^{(1)} = \begin{pmatrix} E_r & O \\ F_3^{(1)} & E_{H^{(1)}-r} \end{pmatrix}$ and $Q_2^{(1)} = \begin{pmatrix} E_r & F_2^{(1)} \\ O & E_{H^{(2)}-r} \end{pmatrix}$ such that

$$Q_1^{(1)} A^{(1)} Q_2^{(1)} = \begin{pmatrix} A_1^{(1)} & O \\ O & C_4^{(1)} \end{pmatrix}.$$

Assume that we have $Q'_1 = \begin{pmatrix} E_r & O \\ F'_3 & E_{H^{(1)}-r} \end{pmatrix}$ and $Q'_2 = \begin{pmatrix} E_r & F'_2 \\ O & E_{H^{(S+1)}-r} \end{pmatrix}$ such that

$$Q'_1 \left(\prod_{s=1}^S A^{(s)} \right) Q'_2 = \begin{pmatrix} C'_1 & O \\ O & \prod_{s=1}^S C^{(s)} \end{pmatrix}.$$

where C'_1 is regular.

Let $A'^{(S+1)} = Q_2'^{-1} A^{(S+1)} = \begin{pmatrix} A'_1 & A'_2 \\ A'_3 & A'_4 \end{pmatrix}$. Then, we have

$$Q'_1 \left(\prod_{s=1}^S A^{(s)} \right) A^{(S+1)} = Q'_1 \left(\prod_{s=1}^S A^{(s)} \right) Q'_2 A'^{(S+1)} = \begin{pmatrix} C'_1 A'_1 & C'_1 A'_2 \\ \prod_{s=1}^S C^{(s)} A'_3 & \prod_{s=1}^S C^{(s)} A'_4 \end{pmatrix}.$$

Because $Q'_1 \begin{pmatrix} E_r & O \\ O & O \end{pmatrix} = \begin{pmatrix} E_r & O \\ F'_3 & O \end{pmatrix}$, we can assume $C'_1 A'_1$ is regular.

Let $Q''_1 = \begin{pmatrix} E_r & O \\ -\prod_{s=1}^S C^{(s)} A'_3 (C'_1 A'_1)^{-1} & E_{h_1-r} \end{pmatrix}$. Then,

$$Q''_1 Q'_1 \left(\prod_{s=1}^S A^{(s)} \right) A^{(S+1)} = \begin{pmatrix} C'_1 A'_1 & C'_1 A'_2 \\ O & -\prod_{s=1}^S C^{(s)} A'_3 A_1'^{-1} A'_2 + \prod_{s=1}^S C^{(s)} A'_4 \end{pmatrix}.$$

We also transform the variables A'_4 to $C^{(S+1)} = -A'_3 A'^{-1}_1 A'_2 + A'_4$, to obtain

$$Q''_1 Q'_1 \left(\prod_{s=1}^S A^{(s)} \right) A^{(S+1)} = \begin{pmatrix} C'_1 A'_1 & C'_1 A'_2 \\ O & \prod_{s=1}^{S+1} C^{(s)} \end{pmatrix}.$$

Let $Q''_2 = \begin{pmatrix} E_r & -A'^{-1}_1 A'_2 \\ O & E_{h_{S+1}-r} \end{pmatrix}$, then we have

$$Q''_1 Q'_1 \left(\prod_{s=1}^S A^{(s)} \right) A^{(S+1)} Q''_2 = \begin{pmatrix} C'_1 A'_1 & O \\ O & \prod_{s=1}^{S+1} C^{(s)} \end{pmatrix}.$$

We transform the variables A'_2 and F'_3 to $F''_2 = -A'^{-1}_1 A'_2$ and $F''_3 = F'_3 -$
115 $\prod_{s=1}^S C^{(s)} A'_3 (C'_1 A'_1)^{-1}$, respectively. Then, we have

$$\begin{aligned} Q''_1 Q'_1 &= \begin{pmatrix} E_r & O \\ F''_3 & E_{h_1-r} \end{pmatrix} \\ Q''_2 &= \begin{pmatrix} E_r & F''_2 \\ O & E_{h_3-r} \end{pmatrix}, \end{aligned}$$

which completes the proof by induction.

End of Proof.

By Theorem 3, we have

$$\begin{aligned} P_1 \left(\prod_{s=1}^L A^{(s)} - \begin{pmatrix} E_r & O \\ O & O \end{pmatrix} \right) P_2 &= \begin{pmatrix} C_1 & O \\ O & \prod_{s=1}^L C^{(s)} \end{pmatrix} - \begin{pmatrix} E_r & F_2 \\ F_3 & F_3 F_2 \end{pmatrix} \\ &= \begin{pmatrix} C_1 - E_r & -F_2 \\ -F_3 & \prod_{s=1}^L C^{(s)} - F_3 F_2 \end{pmatrix}. \end{aligned}$$

Therefore, we have

$$\begin{aligned} \lambda_{w^*} \left\langle \prod_{s=1}^L A^{(s)} - \prod_{s=1}^L A^{*(s)} \right\rangle &= \frac{r^2 + r(H^{(1)} + H^{(L+1)} - 2r)}{2} + \lambda_{w^*} \left\langle \prod_{s=1}^L C^{(s)} \right\rangle \\ &= \frac{-r^2 + r(H^{(1)} + H^{(L+1)})}{2} + \lambda_{w^*} \left\langle \prod_{s=1}^L C^{(s)} \right\rangle. \end{aligned}$$

120 Let $M^{(s)} = H^{(s)} - r$ for $s = 1, \dots, L+1$.

Theorem 4 (Method for determining the deepest singular point [22]).

Let $F_1(w_1, \dots, w_d), \dots, F_m(w_1, \dots, w_d)$ be homogeneous functions of $w_1, \dots, w_j$ ($j \leq d$). Also let φ be a C^∞ function such that $\varphi(0, \dots, 0, w_{j+1}^*, \dots, w_d^*) \geq \varphi(w_1^*, \dots, w_d^*)$ and φ_w is a homogeneous function of $w_1, \dots, w_j$ in a small neighbourhood of $(0, \dots, 0, w_{j+1}^*, \dots, w_d^*)$. Then, we have

$$\lambda_{(0, \dots, 0, w_{j+1}^*, \dots, w_d^*)}(\langle F_1, \dots, F_m \rangle, \varphi) \leq \lambda_{(w_1^*, \dots, w_j^*, w_{j+1}^*, \dots, w_d^*)}(\langle F_1, \dots, F_m \rangle, \varphi).$$

By Theorem 4, we can set $r^{(s)} = r$ for $s = 1, \dots, L$, without loss of generality.

Define for $s = 1, \dots, L$,

$$C^{(s)} = (c_{ij}^{(s)}), \quad (1 \leq i \leq M^{(s)}, 1 \leq j \leq M^{(s+1)}),$$

and consider the log canonical threshold of $\|\prod_{s=1}^L C^{(s)}\|^2$.

We prove the main theorem by developing blow-up method along submanifolds to obtain log canonical thresholds, i.e., learning coefficients. We use a recursive blow-up process.

Define the numerical sequence by

$$M(S) = \min\{M^{(s)} \mid 1 \leq s \leq S\}$$

for $S = 1, \dots, L$. Let

$$T_{S,k} = (t_{S,k}^{(1)}, \dots, t_{S,k}^{(L)})$$

and let $t_{S,k}^{(s)} = M^{(s+1)}$, $\tilde{t}_{S,k} = M(1)$ and $M_{S,k} = 0$ for $1 \leq S, s \leq L, 1 \leq k \leq M(S+1)$ at the first step in this induction proof.

Definition 4. For two vectors $T = (t_1, \dots, t_L)$ and $T' = (t'_1, \dots, t'_L) \in \mathbf{R}^L$, denote

$$T \leq T' \text{ if } t_1 \leq t'_1, \dots, t_L \leq t'_L,$$

and

$$T < T' \text{ if } t_1 \leq t'_1, \dots, t_L \leq t'_L \text{ and } T \neq T'.$$

(Inductive statement)

Let us assume that for $J \geq 0$ and $S \geq 0$

$$\langle \prod_{s=1}^L C^{(s)} \rangle = \langle \text{diag}(b_1, \dots, b_{M(S)}) \begin{pmatrix} E_J & O \\ O & D_J \end{pmatrix} \prod_{s=S+1}^L C^{(s)} \rangle,$$

130 where $\text{diag}(b_1, \dots, b_{M(S)})$ is a diagonal matrix of size $M(S) \times M(S)$.

Let $D_J = (d_{i,j})$ for $J+1 \leq i \leq M(S)$, $J+1 \leq j \leq M^{(S+1)}$ be a matrix of size $(M(S) - J) \times (M^{(S+1)} - J)$ and

$$\begin{aligned} b_0 &= 1 \\ b_i &= \prod_{\tilde{t}_{s,k}=i-1} u_{s,k} b_{i-1}, \quad i = 1, \dots, M(S), \\ \prod_{s=1}^L \prod_{i=1}^{M^{(s)}} \prod_{j=1}^{M^{(s+1)}} dc_{ij}^{(s)} &= \left(\prod_{s=1}^{S-1} \prod_{k=1}^{M^{(s+1)}} u_{s,k}^{M_{s,k}-1} du_{s,k} \right) \left(\prod_{k=1}^J u_{S,k}^{M_{S,k}-1} du_{S,k} \right) \\ &\quad \times \left(\prod_{i=J+1}^{M(S)} \prod_{j=J+1}^{M^{(S+1)}} dd_{ij} \right) \left(\prod_{s=S+1}^L \prod_{i=1}^{M^{(s)}} \prod_{j=1}^{M^{(s+1)}} dc_{ij}^{(s)} \right), \\ \tilde{t}_{s,k} &= \min\{t_{s,k}^{(S')} : 1 \leq S' \leq L\}, \text{ for } 1 \leq s \leq S-1, 1 \leq k \leq M(s+1), \\ \tilde{t}_{S,k} &= \min\{t_{S,k}^{(S')} : 1 \leq S' \leq L\} = t_{S,k}^{(S)}, \text{ for } 1 \leq k \leq J, \end{aligned}$$

by a blow-up process.

Moreover, we have

$$T_{s,k} \leq T_{s',k'}, \text{ or } T_{s,k} \geq T_{s',k'},$$

for $1 \leq s, s' \leq L$, $1 \leq k, k' \leq M(S)$.

135

(End of the inductive statement)

The above is obvious for $S = 0$ and $J = 0$.

Case 1

Assume that

$$b_{J+1} = b_{J+2} = \dots = b_{J+J_1}, b_{J+J_1+1} \neq b_{J+J_1},$$

that is, $\{\tilde{t}_{s,k} \mid \tilde{t}_{s,k} = i\} = \emptyset$ for $i = J+1, \dots, J+J_1-1$.

Fix $u_{s,k}$ such that $\tilde{t}_{s,k} = J+J_1$ and $T_{s,k} \leq T_{s',k'}$ for $\tilde{t}_{s',k'} = J+J_1$.

Construct the blow-up along the submanifold

$$\{d_{ij} = 0(i = J + 1, \dots, J + J_1, j = J + 1, \dots, M^{(S+1)}), u_{s,k} = 0\},$$

140 **Case 1 (1)**

Consider instances in which

$$\begin{pmatrix} d_{J+1,J+1} & d_{J+1,J+2} & \cdots & d_{J+1,M^{(S+1)}} \\ \vdots & \vdots & \cdots & \vdots \\ d_{J+J_1,J+1} & d_{J+J_1,J+2} & \cdots & d_{J+J_1,M^{(S+1)}} \end{pmatrix} \\ = u_{s,k} \begin{pmatrix} d'_{J+1,J+1} & d'_{J+1,J+2} & \cdots & d'_{J+1,M^{(S+1)}} \\ \vdots & \vdots & \cdots & \vdots \\ d'_{J+J_1,J+1} & d'_{J+J_1,J+2} & \cdots & d'_{J+J_1,M^{(S+1)}} \end{pmatrix} \\ \text{Let } \begin{cases} t_{s,k}^{(S)} = t_{s,k}^{(S+1)} = \cdots = t_{s,k}^{(L)} = J, \tilde{t}_{s,k} = J \\ b'_{J+1} = u_{s,k} b_{J+1} \\ \vdots \\ b'_{J+J_1} = u_{s,k} b_{J+J_1} \\ M'_{s,k} = M_{s,k} + J_1(M^{(S+1)} - J), \end{cases}$$

Because $\{\tilde{t}_{s',k'} \mid \tilde{t}_{s',k'} = i\} = \emptyset$ for $i = J + 1, \dots, J + J_1 - 1$, we have

$$T_{s',k'} \leq T_{s'',k''} \text{ or } T_{s',k'} \geq T_{s'',k''},$$

for $1 \leq s', s'' \leq L$ and $1 \leq k', k'' \leq M(S + 1)$.

Again, let $b'_i, M'_{s,k}$ be $b_i, M_{s,k}$, and we have the inductive statement with the

145 number of elements in $\{u_{s',k'} \mid \tilde{t}_{s',k'} = J + J_1\}$ decreased by one.

Case 1 (2)

Consider instances in which

$$\begin{pmatrix} d_{J+1,J+1} & d_{J+1,J+2} & \cdots & d_{J+1,M^{(S+1)}} \\ \vdots & \vdots & \cdots & \vdots \\ d_{J+J_1,J+1} & d_{J+J_1,J+2} & \cdots & d_{J+J_1,M^{(S+1)}} \end{pmatrix} \\ = u_{S,J+1} \begin{pmatrix} 1 & d'_{J+1,J+2} & \cdots & d'_{J+1,M^{(S+1)}} \\ \vdots & \vdots & \cdots & \vdots \\ d'_{J+J_1,J+1} & d'_{J+J_1,J+2} & \cdots & d'_{J+J_1,M^{(S+1)}} \end{pmatrix}$$

and

$$u_{s,k} = u_{S,J+1} u'_{s,k}.$$

$$\text{Let } \begin{cases} t_{S,J+1}^{(i)} = t_{sk}^{(i)}, (i = 1, \dots, S-1) \\ t_{S,J+1}^{(S)} = t_{S,J+1}^{(S+1)} = \dots = t_{S,J+1}^{(L)} = J, \tilde{t}_{S,J+1} = J \\ b'_{J+1} = u_{S,J+1} b_{J+1} \\ \vdots \\ b'_{M(S)} = u_{S,J+1} b_{M(S)} \\ M'_{S,J+1} = M_{sk} + J_1(M^{(S+1)} - J). \end{cases}$$

Because $\{\tilde{t}_{s,k} \mid \tilde{t}_{s,k} = i\} = \phi$ for $i = J+1, \dots, J+J_1-1$, we have

$$T_{s,k} \leq T_{s',k'}, \text{ or } T_{s,k} \geq T_{s',k'},$$

for $1 \leq s, s' \leq L$ and $1 \leq k, k' \leq M(S+1)$.

150

Define regular matrix Q

$$Q = \begin{pmatrix} 1 & -d'_{J+1,J+2} & -d'_{J+1,J+3} & \cdots & -d'_{J+1,M^{(S+1)}} \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & & 1 \end{pmatrix}.$$

and transform the variables introducing

$$D''_J = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ d''_{J+2,J+1} & d''_{J+2,J+2} & \cdots & d''_{J+2,M^{(S+1)}} \\ d''_{J+3,J+1} & d''_{J+3,J+2} & \cdots & d''_{J+3,M^{(S+1)}} \\ \vdots & \vdots & \ddots & \vdots \\ d''_{J+J_1,J+1} & d''_{J+J_1,J+2} & \cdots & d''_{J+J_1,M^{(S+1)}} \\ d''_{J+J_1+1,J+1} & d''_{J+J_1+1,J+2} & \cdots & d''_{J+J_1+1,M^{(S+1)}} \\ \vdots & \vdots & \cdots & \vdots \\ d''_{M(S),J+1} & d''_{M(S),J+2} & \cdots & d''_{M(S),M^{(S+1)}} \end{pmatrix}$$

$$= \begin{pmatrix} 1 & d'_{J+1,J+2} & \cdots & d'_{J+1,M^{(S+1)}} \\ d'_{J+2,J+1} & d'_{J+2,J+2} & \cdots & d'_{J+2,M^{(S+1)}} \\ \vdots & \vdots & \cdots & \vdots \\ d'_{J+J_1,J+1} & d'_{J+J_1,J+2} & \cdots & d'_{J+J_1,M^{(S+1)}} \\ d_{J+J_1+1,J+1} & d_{J+J_1+1,J+2} & \cdots & d_{J+J_1+1,M^{(S+1)}} \\ \vdots & \vdots & \cdots & \vdots \\ d_{M(S),J+1} & d_{M(S),J+2} & \cdots & d_{M(S),M^{(S+1)}} \end{pmatrix} Q.$$

Let $C_J^{(S+1)} = \begin{pmatrix} c_{J+1,1}^{(S+1)} & c_{J+1,2}^{(S+1)} & \cdots & c_{J+1,M^{(S+2)}}^{(S+1)} \\ c_{J+2,1}^{(S+1)} & c_{J+2,2}^{(S+1)} & \cdots & c_{J+2,M^{(S+2)}}^{(S+1)} \\ \vdots & \vdots & \cdots & \vdots \\ c_{M^{(S+1)},1}^{(S+1)} & c_{M^{(S+1)},2}^{(S+1)} & \cdots & c_{M^{(S+1)},M^{(S+2)}}^{(S+1)} \end{pmatrix}$ and transform the variables $C_J^{(S+1)}$ to $C_J'^{(S+1)}$ using $C_J'^{(S+1)} = Q^{-1}C_J^{(S+1)}$.

Define regular matrix P

$$P = \begin{pmatrix} 1 & 0 & 0 & 0 & \cdots & 0 \\ -\frac{b'_{J+2}}{b'_{J+1}}d''_{J+2,J+1} & 1 & 0 & 0 & \cdots & 0 \\ -\frac{b'_{J+3}}{b'_{J+1}}d''_{J+3,J+1} & 0 & 1 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ -\frac{b'_{M(S)}}{b'_{J+1}}d''_{M(S),J+1} & 0 & \cdots & \cdots & \cdots & 1 \end{pmatrix}.$$

and let

$$D_J''' = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & d''_{J+2,J+2} & \cdots & d''_{J+2,M^{(S+1)}} \\ 0 & d''_{J+3,J+2} & \cdots & d''_{J+3,M^{(S+1)}} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & d''_{M(S),J+2} & \cdots & d''_{M(S),M^{(S+1)}} \end{pmatrix} = \begin{pmatrix} 1 & O \\ O & D_{J+1} \end{pmatrix}.$$

155 We then have

$$\begin{aligned} & P \text{diag}(b_{J+1}, \dots, b_{M(S)}) D_J C_J^{(S+1)} \\ &= P u_{S,J+1} \text{diag}(b'_{J+1}, \dots, b'_{M(S)}) D_J'' C_J'^{(S+1)} \end{aligned}$$

$$\begin{aligned}
&= u_{S,J+1} P \text{diag}(b'_{J+1}, \dots, b'_{M(S)}) D_J'' C_J'^{(S+1)} \\
&= u_{S,J+1} \text{diag}(b'_{J+1}, \dots, b'_{M(S)}) D_J''' C_J'^{(S+1)}.
\end{aligned}$$

Again, let $b'_i, M'_{sk}, d'''_{ij}, c'_{ij}$ be $b_i, M_{sk}, d_{ij}, c_{ij}$.

If $J+1 \leq M(S+1) = \min\{M(S), M^{(S+1)}\}$, then we have the inductive statement with J increased by one.

If $J+1 > M(S+1) = \min\{M(S), M^{(S+1)}\}$, then $D_J''' = (1, 0, \dots, 0)$ or $D_J''' = (1, 0, \dots, 0)^t$ with t the operation of transposition. Therefore, we have

$$\left\langle \prod_{s=1}^L C^{(s)} \right\rangle = \langle \text{diag}(b_1, \dots, b_J, b_{J+1}, \dots, b_{M(S+1)}) C'^{(S+1)} \prod_{s=S+2}^L C^{(s)} \rangle,$$

$$\text{where } C'^{(S+1)} = \begin{pmatrix} c_{1,1}^{(S+1)} & c_{1,2}^{(S+1)} & \dots & c_{1,M^{(S+2)}}^{(S+1)} \\ c_{2,1}^{(S+1)} & c_{2,2}^{(S+1)} & \dots & c_{2,M^{(S+2)}}^{(S+1)} \\ \vdots & \vdots & \vdots & \vdots \\ c_{J,1}^{(S+1)} & c_{J,2}^{(S+1)} & \dots & c_{J,M^{(S+2)}}^{(S+1)} \\ c_{J+1,1}^{(S+1)} & c_{J+1,2}^{(S+1)} & \dots & c_{J+1,M^{(S+2)}}^{(S+1)} \\ \vdots & \vdots & \vdots & \vdots \\ c_{M(S+1),1}^{(S+1)} & c_{M(S+1),2}^{(S+1)} & \dots & c_{M(S+1),M^{(S+2)}}^{(S+1)} \end{pmatrix}, \text{ which is}$$

the inductive statement with S increased by one.

Case 2

Assume that

$$b_{J+1} = b_{J+2} = \dots = b_{M(S)},$$

that is, $\{\tilde{t}_{s,k} \mid \tilde{t}_{s,k} = i\} = \phi$ for $i = J+1, \dots, M(S) - 1$.

Construct the blow-up along submanifold

$$\{d_{ij} = 0, (i = J+1, \dots, M(S), j = J+1, \dots, M^{(S+1)})\},$$

Consider instances in which

$$\begin{pmatrix} d_{J+1,J+1} & d_{J+1,J+2} & \dots & d_{J+1,M^{(S+1)}} \\ \vdots & \vdots & \dots & \vdots \\ d_{M(S),J+1} & d_{M(S),J+2} & \dots & d_{M(S),M^{(S+1)}} \end{pmatrix}$$

$$= u_{S,J+1} \begin{pmatrix} 1 & d'_{J+1,J+2} & \cdots & d'_{J+1,M^{(S+1)}} \\ \vdots & \vdots & \cdots & \vdots \\ d'_{M^{(S)},J+1} & d'_{M^{(S)},J+2} & \cdots & d'_{M^{(S)},M^{(S+1)}} \end{pmatrix}.$$

$$\text{Let } \begin{cases} t_{S,J+1}^{(i)} = M^{(i+1)}, (i = 1, \dots, S-1) \\ t_{S,J+1}^{(S)} = t_{S,J+1}^{(S+1)} = \cdots = t_{S,J+1}^{(L)} = J, \tilde{t}_{S,J+1} = J \\ b'_{J+1} = u_{S,J+1} b_{J+1} \\ \vdots \\ b'_{M^{(S)}} = u_{S,J+1} b_{M^{(S)}} \\ M'_{S,J+1} = (M^{(S)} - J)(M^{(S+1)} - J). \end{cases}$$

Because $\{\tilde{t}_{s,k} \mid \tilde{t}_{s,k} = i\} = \emptyset$ for $i = J+1, \dots, M^{(S)} - 1$, we have

$$T_{s,k} \leq T_{s',k'}, \text{ or } T_{s,k} \geq T_{s',k'},$$

for $1 \leq s, s' \leq L$ and $1 \leq k, k' \leq M^{(S+1)}$.

Define regular matrix Q

$$Q = \begin{pmatrix} 1 & -d'_{J+1,J+2} & -d'_{J+1,J+3} & \cdots & -d'_{J+1,M^{(S+1)}} \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & & 1 \end{pmatrix}.$$

and transform the variables using

$$\begin{aligned} D''_J &= \begin{pmatrix} 1 & 0 & \cdots & 0 \\ d''_{J+2,J+1} & d''_{J+2,J+2} & \cdots & d''_{J+2,M^{(S+1)}} \\ d''_{J+3,J+1} & d''_{J+3,J+2} & \cdots & d''_{J+3,M^{(S+1)}} \\ \vdots & \vdots & \ddots & \vdots \\ d''_{M^{(S)},J+1} & d''_{M^{(S)},J+2} & \cdots & d''_{M^{(S)},M^{(S+1)}} \end{pmatrix} \\ &= \begin{pmatrix} 1 & d'_{J+1,J+2} & \cdots & d'_{J+1,M^{(S+1)}} \\ d'_{J+2,J+1} & d'_{J+2,J+2} & \cdots & d'_{J+2,M^{(S+1)}} \\ \vdots & \vdots & \cdots & \vdots \\ d'_{M^{(S)},J+1} & d'_{M^{(S)},J+2} & \cdots & d'_{M^{(S)},M^{(S+1)}} \end{pmatrix} Q. \end{aligned}$$

Let $C_J^{(S+1)} = \begin{pmatrix} c_{J+1,1}^{(S+1)} & c_{J+1,2}^{(S+1)} & \cdots & c_{J+1,M^{(S+2)}}^{(S+1)} \\ c_{J+2,1}^{(S+1)} & c_{J+2,2}^{(S+1)} & \cdots & c_{J+2,M^{(S+2)}}^{(S+1)} \\ \vdots & \vdots & \cdots & \vdots \\ c_{M^{(S+1)},1}^{(S+1)} & c_{M^{(S+1)},2}^{(S+1)} & \cdots & c_{M^{(S+1)},M^{(S+2)}}^{(S+1)} \end{pmatrix}$ and transform the variables $C_J^{(S+1)}$ to $C_J'^{(S+1)}$ using $C_J'^{(S+1)} = Q^{-1}C_J^{(S+1)}$.

170

Define regular matrix P

$$P = \begin{pmatrix} 1 & 0 & 0 & 0 & \cdots & 0 \\ -\frac{b'_{J+2}}{b'_{J+1}}d''_{J+2,J+1} & 1 & 0 & 0 & \cdots & 0 \\ -\frac{b'_{J+3}}{b'_{J+1}}d''_{J+3,J+1} & 0 & 1 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ -\frac{b'_{M(S)}}{b'_{J+1}}d''_{M(S),J+1} & 0 & \cdots & \cdots & \cdots & 1 \end{pmatrix}.$$

and let

$$D_J''' = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & d''_{J+2,J+2} & \cdots & d''_{J+2,M^{(S+1)}} \\ 0 & d''_{J+3,J+2} & \cdots & d''_{J+3,M^{(S+1)}} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & d''_{M(S),J+2} & \cdots & d''_{M(S),M^{(S+1)}} \end{pmatrix} = \begin{pmatrix} 1 & O \\ O & D_{J+1} \end{pmatrix}.$$

We then have

$$\begin{aligned} & P \text{diag}(b_{J+1}, \dots, b_{M(S)}) D_J C_J^{(S+1)} \\ &= P u_{S,J+1} \text{diag}(b'_{J+1}, \dots, b'_{M(S)}) D_J'' C_J'^{(S+1)} \\ &= u_{S,J+1} P \text{diag}(b'_{J+1}, \dots, b'_{M(S)}) D_J'' C_J'^{(S+1)} \\ &= u_{S,J+1} \text{diag}(b'_{J+1}, \dots, b'_{M(S)}) D_J'' C_J'^{(S+1)}. \end{aligned}$$

Again, let $b'_{ij}, M'_{ij}, d''_{ij}, c'_{ij}$ be $b_{ij}, M_{ij}, d_{ij}, c_{ij}$.

If $J+1 \leq M(S+1) = \min\{M(S), M^{(S+1)}\}$, then we have the inductive statement with J increased by one.

If $J+1 > M(S+1) = \min\{M(S), M^{(S+1)}\}$, then $D_J''' = (1, 0, \dots, 0)$ or

$D_J''' = (1, 0, \dots, 0)^t$. Therefore, we have

$$\langle \prod_{s=1}^L C^{(s)} \rangle = \langle \text{diag}(b_1, \dots, b_J, b'_{J+1}, \dots, b'_{M(S+1)}) C'^{(S+1)} \prod_{s=S+2}^L C^{(s)} \rangle,$$

175 where $C'^{(S+1)} = \begin{pmatrix} c_{1,1}^{(S+1)} & c_{1,2}^{(S+1)} & \dots & c_{1,M(S+2)}^{(S+1)} \\ c_{2,1}^{(S+1)} & c_{2,2}^{(S+1)} & \dots & c_{2,M(S+2)}^{(S+1)} \\ \vdots & \vdots & \vdots & \vdots \\ c_{M(S+1),1}^{(S+1)} & c_{M(S+1),2}^{(S+1)} & \dots & c_{M(S+1),M(S+2)}^{(S+1)} \end{pmatrix}$, which is the inductive statement with S increased by one.

We finally have the inductive statement for $S = L + 1$,

$$\langle \prod_{s=1}^L C^{(s)} \rangle = \langle \text{diag}(b_1, \dots, b_{M(L+1)}) \rangle.$$

Candidates for the log canonical threshold of $\|\prod_{s=1}^L C^{(s)}\|^2$ on this local coordinate are

$$\frac{1}{2} \min\{M_{s,k}, \tilde{t}_{s,k} = 0\}$$

and

$$M_{s,k} = (M^{(1)} - t_{s,k}^{(1)})(M^{(2)} - t_{s,k}^{(1)}) + \sum_{j=2}^L (t_{s,k}^{(j-1)} - t_{s,k}^{(j)})(M^{(j+1)} - t_{s,k}^{(j)}).$$

Define ℓ , H_i and S_j such that

$$H_i \in \{t_{s,k}^{(1)}, \dots, t_{s,k}^{(L)}\},$$

$$M_{s,k} = (M^{(S_1)} - H_1)(M^{(S_2)} - H_1) + \sum_{j=2}^{\ell} (H_{j-1} - H_j)(M^{(S_{j+1})} - H_j),$$

where $S_j < S_{j+1}$, $H_j \leq H_{j-1}$, $H_j \leq M^{(S_{j+1})}$ and $H_\ell = 0$. For every $T_{s,k}$, we have such sequences (H_i) and (S_i) .

Let $F_j = H_{j-1} - H_j + M^{(S_{j+1})}$ for $j = 2, \dots, \ell$ and $F_1 = M^{(S_1)} + M^{(S_2)} - H_1$.

Then, we have

$$H_j = H_{j-1} - F_j + M^{(S_{j+1})} = H_1 - \sum_{l=2}^j (F_l - M^{(S_{l+1})}) = -\sum_{l=1}^j F_l + \sum_{l=1}^{j+1} M^{(S_l)}.$$

Because $H_{j-1} - H_j \geq 0$, $M^{(S_{j+1})} - H_j \geq 0$ and $H_\ell = 0$, we have

$$\begin{aligned} F_j - M^{(S_{j+1})} &\geq 0, \\ M^{(S_{j+1})} - H_j &= \sum_{l=1}^j F_l - \sum_{l=1}^j M^{(S_l)} \geq 0, \\ H_\ell &= -\sum_{l=1}^{\ell} F_l + \sum_{l=1}^{\ell+1} M^{(S_l)} = 0. \end{aligned}$$

180

Also we have

$$F_\ell - M^{(S_{\ell+1})} = -\sum_{l=1}^{\ell-1} F_l + \sum_{l=1}^{\ell} M^{(S_l)} \geq 0,$$

and

$$\begin{aligned} M_{s,k} &= (F_1 - M^{(S_2)})(F_1 - M^{(S_1)}) + \sum_{j=2}^{\ell} (F_j - M^{(S_{j+1})}) \left(\sum_{l=1}^j F_l - \sum_{l=1}^j M^{(S_l)} \right) \\ &= \sum_{j=1}^{\ell-1} (F_j - M^{(S_{j+1})}) \left(\sum_{l=1}^j F_l - \sum_{l=1}^j M^{(S_l)} \right) + \left(-\sum_{j=1}^{\ell-1} F_j + \sum_{j=1}^{\ell} M^{(S_j)} \right) M^{(S_{\ell+1})} \\ &= \sum_{j=1}^{\ell-1} F_j^2 - \left(\sum_{k=1}^{\ell+1} M^{(S_k)} \right) \left(\sum_{j=1}^{\ell-1} F_j \right) + \sum_{1 \leq i < j \leq \ell-1} F_i F_j + \sum_{1 \leq i < j \leq \ell+1} M^{(S_i)} M^{(S_j)} \\ &= \sum_{j=1}^{\ell-1} \left(F_j - \frac{\sum_{k=1}^{\ell+1} M^{(S_k)}}{\ell} \right)^2 + \sum_{j=1}^{\ell-1} \frac{2 \sum_{k=1}^{\ell+1} M^{(S_k)}}{\ell} F_j - (\ell-1) \left(\frac{\sum_{k=1}^{\ell+1} M^{(S_k)}}{\ell} \right)^2 \\ &\quad - \left(\sum_{k=1}^{\ell+1} M^{(S_k)} \right) \sum_{j=1}^{\ell-1} F_j + \sum_{1 \leq i < j \leq \ell-1} \left(F_i - \frac{\sum_{k=1}^{\ell+1} M^{(S_k)}}{\ell} \right) \left(F_j - \frac{\sum_{k=1}^{\ell+1} M^{(S_k)}}{\ell} \right) \\ &\quad + \frac{(\ell-2) \sum_{k=1}^{\ell+1} M^{(S_k)}}{\ell} \sum_{j=1}^{\ell-1} F_j - \frac{(\ell-1)(\ell-2)}{2} \left(\frac{\sum_{k=1}^{\ell+1} M^{(S_k)}}{\ell} \right)^2 \\ &\quad + \sum_{1 \leq i < j \leq \ell+1} M^{(S_i)} M^{(S_j)} \\ &= \sum_{j=1}^{\ell-1} \left(F_j - \frac{\sum_{k=1}^{\ell+1} M^{(S_k)}}{\ell} \right)^2 + \sum_{1 \leq i < j \leq \ell-1} \left(F_i - \frac{\sum_{k=1}^{\ell+1} M^{(S_k)}}{\ell} \right) \left(F_j - \frac{\sum_{k=1}^{\ell+1} M^{(S_k)}}{\ell} \right) \\ &\quad - \frac{\ell(\ell-1)}{2} \left(\frac{\sum_{k=1}^{\ell+1} M^{(S_k)}}{\ell} \right)^2 + \sum_{1 \leq i < j \leq \ell+1} M^{(S_i)} M^{(S_j)} \\ &= \frac{1}{2} \sum_{j=1}^{\ell-1} \left(F_j - \frac{\sum_{k=1}^{\ell+1} M^{(S_k)}}{\ell} \right)^2 + \frac{1}{2} \left(\sum_{j=1}^{\ell-1} F_j - \frac{\ell-1}{\ell} \sum_{j=1}^{\ell+1} M^{(S_j)} \right)^2 \end{aligned}$$

$$-\frac{\ell(\ell-1)}{2}\left(\frac{\sum_{k=1}^{\ell+1} M^{(S_k)}}{\ell}\right)^2 + \sum_{1 \leq i < j \leq \ell+1} M^{(S_i)} M^{(S_j)}.$$

Lemma 3. Let ℓ and $0 \leq a, b \leq \ell-1$ be natural numbers and let

$$\frac{A(b)}{\ell^2} = b\left(\frac{\ell-a}{\ell}\right)^2 + (\ell-1-b)\left(-\frac{a}{\ell}\right)^2 + \left(b\frac{\ell-a}{\ell} - (\ell-1-b)\frac{a}{\ell}\right)^2.$$

Then,

$$\min\{A|b = 0, \dots, \ell-1\} = A(a-1) = A(a) = a\ell(\ell-a).$$

(Proof)

We have

$$\frac{A(b)}{\ell^2} = \frac{b(\ell-a)^2 + (\ell-1-b)a^2 + (b(\ell-a) - (\ell-1-b)a)^2}{\ell^2}.$$

and

$$\frac{\partial A(b)}{\partial b} = (\ell-a)^2 - a^2 + 2(b(\ell-a) - (\ell-1-b)a)\{(\ell-a) + a\} = \ell^2(1+2b-2a).$$

185 Therefore, we have established the proof.

End of Proof.

Define ℓ , $\mathcal{M}$, M , S_j and a as in Definition 3.

By Lemma 3, we have

$$\begin{aligned} & 2\lambda_O(\|\prod_{s=1}^L C^{(s)}\|^2) \\ &= \frac{a(\ell-a)}{2\ell} - \frac{\ell(\ell-1)}{2}\left(\frac{\sum_{j=1}^{\ell+1} M^{(S_j)}}{\ell}\right)^2 + \sum_{1 \leq i < j \leq \ell+1} M^{(S_i)} M^{(S_j)} \\ &= \frac{a(\ell-a)}{2\ell} - \frac{\ell(\ell-1)}{2}\left(M + \frac{a-\ell}{\ell}\right)^2 + \sum_{1 \leq i < j \leq \ell+1} M^{(S_i)} M^{(S_j)} \\ &= \frac{a(\ell-a)}{2\ell} - \frac{\ell(\ell-1)}{2}\left(M^2 + 2\frac{a-\ell}{\ell}M + \frac{(a-\ell)^2}{\ell^2}\right) + \sum_{1 \leq i < j \leq \ell+1} M^{(S_i)} M^{(S_j)} \\ &= -\frac{1}{2}(\ell-a-1)(\ell-a) - \frac{\ell(\ell-1)}{2}\left(M^2 + 2\frac{a-\ell}{\ell}M\right) + \sum_{1 \leq i < j \leq \ell+1} M^{(S_i)} M^{(S_j)} \end{aligned}$$

Let

$$\begin{aligned}\tilde{H}_j &= \sum_{l=1}^{j+1} M^{(S_l)} - jM, \text{ for } j = 1, \dots, a, \\ \tilde{H}_j &= \sum_{l=1}^{j+1} M^{(S_l)} - aM - (j-a)(M-1), \text{ for } j = a+1, \dots, \ell,\end{aligned}$$

and

$$\begin{aligned}\tilde{H}'_j &= \sum_{l=1}^{j+1} M^{(S_l)} - j(M-1), \text{ for } j = 1, \dots, \ell-a, \\ \tilde{H}'_j &= \sum_{l=1}^{j+1} M^{(S_l)} - (\ell-a)(M-1) - (j-\ell+a)M \text{ for } j = \ell-a+1, \dots, \ell.\end{aligned}$$

Then, we have $\tilde{H}_\ell = \tilde{H}'_\ell = 0$. Moreover, let $\tilde{T}$ and $\tilde{T}'$ be the vectors corresponding to $(\tilde{H}_j)$ and $(\tilde{H}'_j)$, respectively. That i) $\tilde{T}$ and $\tilde{T}'$ correspond to λ , and ii) $\tilde{T}$ has minimum one and $\tilde{T}'$ has maximum one, are obvious from Definition 4.

For obtaining the values θ , we prepare the following Lemma 4.

Lemma 4. *If the vector $T_{sk} = (t_{sk}^{(1)}, \dots, t_{sk}^{(L)})$ satisfies*

$$\tilde{T} \leq T_{sk} \leq \tilde{T}'$$

and if ℓ , (H_j) , (S_j) which corresponds to T_{sk} satisfies $H_{j-1} - H_j + M^{(S_{j+1})} = t_{sk}^{(S_j-1)} - t_{sk}^{(S_{j+1}-1)} + M^{(S_{j+1})} = M-1$ or M , then, T_{sk} corresponds to λ .

(Proof)

Because $\tilde{T} \leq T_{sk} \leq \tilde{T}'$ and $\tilde{H}_\ell = \tilde{H}'_\ell = 0$, we have $H_\ell = 0$. Therefore, the number of elements of $\{j : H_{j-1} - H_j + M^{(S_{j+1})} = M\}$ is a and that of $\{j : H_{j-1} - H_j + M^{(S_{j+1})} = M-1\}$ is $\ell-a$.

End of Proof.

Lemma 5. *We have*

$$\theta = a(\ell-a) + 1.$$

(Proof)

Given $t_{s,k}^{(S_{j+1}-1)} = H_j$, then by Lemma 4, the number of vectors $T_{s,k}$ corresponding to λ is less than the sum of the number of the set $\cup_{j=1}^{\ell-1} \{H : \tilde{H}_j \leq H \leq \tilde{H}'_j\}$. We have the number of the set $\{H : \tilde{H}_j \leq H \leq \tilde{H}'_j\}$ as

$$\begin{aligned} j+1, & \quad \text{if } j \leq \min\{a, \ell-a\} \\ \min\{a, \ell-a\}+1, & \quad \text{if } \min\{a, \ell-a\}+1 \leq j \leq \max\{a, \ell-a\} \\ \min\{a, \ell-a\}+1+\max\{a, \ell-a\}-j, & \quad \text{if } \max\{a, \ell-a\}+1 \leq j \leq \ell. \end{aligned}$$

Because J is increased by one for Case 1(2) in the proof, we have $\theta \leq a(\ell-a)+1$. Finally, by using T_{sk} , we construct the local coordinate for $\theta = a(\ell-a)+1$ as follows.

Define

$$t_{s,k}^{(S)} = \begin{cases} M(S+1), & S < s, \\ k-1, & S = s, \\ 0, & S > s, \end{cases}$$

for $s < S_2 - 1$,

$$t_{s,k}^{(S)} = \begin{cases} M(S+1), & S < S_2 - 1, \\ \tilde{H}_{j-1}, & S_j - 1 \leq S < S_{j+1} - 1, j < j_0 \\ \tilde{H}_{j_0-1}, & S_{j_0} - 1 \leq S < s, \\ k-1, & S = s, \\ 0, & S > s, \end{cases} \quad (1)$$

for $S_{j_0} - 1 \leq s < S_{j_0+1} - 1, k < \tilde{H}_{j_0-1} + 1, j_0 \geq 2$,

$$t_{s,k}^{(S)} = \begin{cases} M(S+1), & S < s, \\ k-1, & S = s, \\ \tilde{H}'_{j_0-1}, & s < S < S_{j_0+1} - 1, \\ \tilde{H}'_{j-1}, & S_j - 1 \leq S < S_{j+1} - 1, j > j_0, \end{cases} \quad (2)$$

for $S_{j_0} - 1 \leq s < S_{j_0+1} - 1, k \geq \tilde{H}'_{j_0-1} + 1, j_0 \geq 2, (s, k) \neq (S_2 - 1, \tilde{H}'_1 + 1)$,

$$t_{s,k}^{(S)} = \begin{cases} M(S+1), & S < s, \\ \tilde{H}'_{j-1}, & S_j - 1 \leq S < S_{j+1} - 1, 2 \leq j \leq \ell - a + 1 \\ \tilde{H}'_{\ell-a+1} + 1, & S = S_{\ell-a+2} - 1 \\ \tilde{H}'_{j-1}, & S > S_{\ell-a+2} - 1, S_j - 1 \leq S < S_{j+1} - 1, \end{cases} \quad (3)$$

for $s = S_2 - 1, k = \tilde{H}'_1 + 1$,

$$t_{s,k}^{(S)} = \begin{cases} M(S+1), & S < S_2 - 1, \\ \tilde{H}'_{j-1} - j + 1, & S_j - 1 \leq S < S_{j+1} - 1, 2 \leq j \leq j_0 + 1 \\ \tilde{H}'_{j-1} - j_0, & S_j - 1 \leq S < S_{j+1} - 1, j_0 + 1 < j \leq j_0 + \ell - a + 1 \\ \tilde{H}'_{j-1} - j_0 + 1, & S = S_{j+1} - 1, j = j_0 + \ell - a + 1 \\ \tilde{H}'_{j-1}, & S > S_{j_0+\ell-a+2} - 1, S_j - 1 \leq S < S_{j+1} - 1, \end{cases} \quad (4)$$

for $s = S_{j_0+1} - 1, k = \tilde{H}_{j_0} + 1, j_0 \leq a$ and

$$t_{s,k}^{(S)} = \begin{cases} M(S+1), & S < S_2 - 1, \\ \tilde{H}'_{j-1} - j + 1, & S_j - 1 \leq S < S_{j+1} - 1, 2 \leq j \leq \alpha - 1, \\ \tilde{H}'_{j-1} - \alpha + 1, & S_j - 1 \leq S < S_{j+1} - 1, \alpha \leq j \leq j_0, \\ \tilde{H}'_{j-1} - \alpha + 1 - (j - j_0), & S_j - 1 \leq S < S_{j+1} - 1, j_0 < j \leq j_0 + (a - \alpha) + 1, \\ \tilde{H}_{j-1}, & S \geq S_{j_0+(a-\alpha)+2} - 1, S_j - 1 \leq S < S_{j+1} - 1, \end{cases} \quad (5)$$

for $\alpha = \tilde{H}'_{j_0} + 1 - k, S_{j_0+1} - 1 \leq s < S_{j_0+2} - 1, \tilde{H}_{j_0} + 1 \leq k < \tilde{H}'_{j_0} + 1, j_0 > \alpha$.

215 We construct the blow-up process with $T_{s,k}$ in Eqs. (3) and (4) in **Case 1** (2). Thus, the proof is established.

End of Proof.

6. Conclusions

220 We have determined the exact values for the learning coefficients of multiple-layered neural networks with linear units, which generalizes the result of paper [12]. We use the inductive method and a recursive blow-up method and obtain a manifold with a resolution map of singularities.

The main theorems imply that even though model complexities are increased, generalization errors which are expressed by the learning coefficients λ are decreased when the number of layers is increased. This fact seems to be one of the reasons for multiple-layered neural networks having better efficiencies for learning. For example, if the matrix sizes are all the same $H^{(1)} \times H^{(1)}$ then

$$\begin{aligned}\lambda &= \frac{-r^2 + 2rH^{(1)}}{2} + \frac{a(L-a)}{4L} + \frac{L+1}{4L}(H^{(1)}-r)^2 \\ &\rightarrow \frac{-r^2 + 2rH^{(1)}}{2} + \frac{H^{(1)}-r}{4} + \frac{1}{4}(H^{(1)}-r)^2\end{aligned}$$

if $L \rightarrow \infty$, where r denotes the rank of a true probability density function, L the number of layers, and $a = L(\frac{L+1}{L}(H^{(1)}-r) - \lceil \frac{L+1}{L}(H^{(1)}-r) \rceil + 1)$; here $\lceil x \rceil$ denotes the ceiling function of the real number x . On the other hand, note that its order $\theta = a(L-a) + 1 \rightarrow \infty$ as $L \rightarrow \infty$.

Recently, these theoretical values of the learning coefficients have been used effectively in numerical experiments such as information criteria, the Markov chain Monte Carlo [33, 34], and model selection methods. We shall also consider these applications in the future.

Acknowledgments

This research was funded by the Ministry of Education, Culture, Sports, Science, and Technology in Japan, Grant-in-Aid for Scientific Research 18K11479. We thank Richard Haase, PhD, from Edanz (<https://jp.edanz.com/ac>) for editing a draft of this manuscript.

References

- [1] S. Watanabe, Algebraic Geometry and Statistical Learning Theory, Vol. 25, Cambridge University Press, New York, USA, 2009.
- [2] H. Akaike, A new look at the statistical model identification, IEEE Trans. on Automatic Control 19 (1974) 716–723.

- [3] K. Takeuchi, Distribution of an information statistic and the criterion for the optimal model, *Mathematical Science* 153 (1976) 12–18.
- [4] E. J. Hannan, B. G. Quinn, The determination of the order of an autoregression, *Journal of Royal Statistical Society Series B* 41 (1979) 190–195.
- 250 [5] N. J. Murata, S. G. Yoshizawa, S. Amari, Network information criterion - determining the number of hidden units for an artificial neural network model, *IEEE Trans. on Neural Networks* 5 (6) (1994) 865–872.
- [6] G. Schwarz, Estimating the dimension of a model, *Annals of Statistics* 6 (2) (1978) 461–464.
- 255 [7] J. Rissanen, Universal coding, information, prediction, and estimation, *IEEE Trans. on Information Theory* 30 (4) (1984) 629–636.
- [8] S. Watanabe, Algebraic analysis for nonidentifiable learning machines, *Neural Computation* 13 (4) (2001a) 899–933.
- [9] S. Watanabe, Algebraic geometrical methods for hierarchical learning machines, *Neural Networks* 14 (8) (2001b) 1049–1060.
- 260 [10] S. Watanabe, Equations of states in singular statistical estimation, *Neural Networks* 23 (1) (2010) 20–34.
- [11] S. Watanabe, *Mathematical Theory of Bayesian Statistics*, New York, NY, USA, 2018.
- 265 [12] M. Aoyagi, S. Watanabe, Stochastic complexities of reduced rank regression in Bayesian estimation, *Neural Networks* 18 (2005b) 924–933.
- [13] M. Drton, Conference lecture: Bayesian information criterion for singular models, *Algebraic Statistics 2012 in the Alleghenies at The Pennsylvania State University*, <http://jasonmorton.com/aspsu2012/> (2012).
- 270 [14] S. Watanabe, Algebraic geometry of learning machines with singularities and their prior distributions, *Journal of Japanese Society of Artificial Intelligence* 16 (2) (2001c) 308–315.

- [15] H. Hironaka, Resolution of singularities of an algebraic variety over a field of characteristic zero, *Annals of Math* 79 (1964) 109–326.
- 275 [16] W. Fulton, Introduction to toric varieties, *Annals of Mathematics Studies*, Princeton University Press, Princeton, NJ, USA, 1993.
- [17] J. Kollár, Singularities of pairs, *Algebraic geometry-Santa Cruz 1995*, *Proc. Symp. Pure Math.*, American Mathematical Society, Providence, RI, 62 (1997) 221–287.
- 280 [18] M. Mustata, Singularities of pairs via jet schemes, *J. Amer. Math. Soc.* 15 (2002) 599–615.
- [19] M. Kashiwara, B-functions and holonomic systems, *Inventiones Math.* 38 (1976) 33–53.
- [20] M. Aoyagi, S. Watanabe, Resolution of singularities and the generalization error with Bayesian estimation for layered neural network, *IEICE Trans.* J88-D-II 10 (2005a) 2112–2124.
- 285 [21] M. Aoyagi, The zeta function of learning theory and generalization error of three layered neural perceptron, *RIMS Kokyuroku*, Recent Topics on Real and Complex Singularities 1501 (2006) 153–167.
- 290 [22] M. Aoyagi, Consideration on singularities in learning theory and the learning coefficient, *Entropy* 15 (9) (2013) 3714–3733.
- [23] M. Aoyagi, Learning coefficient of Vandermonde matrix-type singularities in model selection, *Entropy (Information Theory, Probability and Statistics)* 21 (6-561) (2019) 1–12.
- 295 [24] M. Aoyagi, Learning coefficients and information criteria, *Frontiers in Artificial Intelligence and Applications* (2019) 351–362.
- [25] M. Aoyagi, Learning coefficient in Bayesian estimation of restricted Boltzmann machine, *Journal of Algebraic Statistics* 4 (1) (2013) 30–57.

- [26] D. Rusakov, D. Geiger, Asymptotic model selection for naive Bayesian
300 networks, Proceedings of the Eighteenth Conference on Uncertainty in Artificial Intelligence (2002) 438–445.
- [27] D. Rusakov, D. Geiger, Asymptotic model selection for naive Bayesian networks, Journal of Machine Learning Research 6 (2005) 1–35.
- [28] P. Zwiernik, An asymptotic behavior of the marginal likelihood for general
305 Markov models, Journal of Machine Learning Research 12 (2011) 3283–3310.
- [29] M. Drton, S. Lin, L. Weihs, P. Zwiernik, Marginal likelihood and model selection for Gaussian latent tree and forest models, Bernoulli 23 (2) (2017) 1202–1232.
- [30] M. Aoyagi, Log canonical threshold of Vandermonde matrix type singularities and generalization error of a three layered neural network, International
310 Journal of Pure and Applied Mathematics 52 (2) (2009) 177–204.
- [31] M. Aoyagi, A Bayesian learning coefficient of generalization error and Vandermonde matrix-type singularities, Communications in Statistics - Theory
315 and Methods 39 (15) (2010) 2667–2687.
- [32] S. Lin, Asymptotic approximation of marginal likelihood integrals, (preprint) (2010).
- [33] K. Nagata, S. Watanabe, Exchange Monte Carlo sampling from Bayesian posterior for singular learning machines, IEEE Transactions on Neural Networks
320 19 (7) (2008a) 1253–1266.
- [34] K. Nagata, S. Watanabe, Asymptotic behavior of exchange ratio in exchange Monte Carlo method, International Journal of Neural Networks 21 (7) (2008b) 980–988.